

Post-Quantum Security Evaluation of Aeronautical Communications

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Abstract—The L-Band Digital Aeronautical Communication System (LDACS) aims to transform aviation communications into a quantum-safe setup, where recent proposals utilize Physical Unclonable Functions (PUFs) as an authentication mechanism and quantum-safe algorithms as a Key Encapsulation Mechanism (KEM). However, the existence of novel techniques that predict the behavior of PUFs, along with severe vulnerabilities in Post-Quantum Cryptography (PQC) schemes being exploitable by attacks running on classical computing infrastructures, raises concerns about their security. In this context, the paper performs a security assessment of LDACS and the extent to which the secret quantities used during the mutual authentication between an aircraft and ground control can be disclosed. It is shown that an intelligent quantum preimage algorithm can reveal these aforementioned PUF secret quantities, which are also susceptible to sophisticated side channel attacks even on isolated workstations. In addition, information set decoding and entropy are used to evaluate the strength of the proposed KEMs for LDACS, which are also benchmarked to identify trade-offs between security and performance. Overall, our findings suggest a hardware implementation of CRYSTALS-Kyber as KEM, to further reduce latency, and the utilization of a public key infrastructure for authentication.

Index Terms—Post-quantum cryptography; Key encapsulation mechanism; Physical unclonable functions; Aviation; LDACS.

I. INTRODUCTION

In the early stages of aviation, the inbound and outbound flights were effectively managed by air traffic controllers due to the manageable number of flights. The evolution of radio communications made Air Traffic Management (ATM) more effective, but significant advances were brought with the adoption of on-board radars and radio-augmented navigation systems. Those systems improved autonomy and precision, considerably increasing the safety of flights.

Data Link (DL) is a type of communication in civilian and military aeronautical operations. It is a digital communication method that is used to transmit Air-to-Air (A/A) or Air-to-Ground (A/G) signals. Unlike traditional voice communications, which are based on analog signals, the DL enhances efficiency, safety, and reliability because it contains additional data such as the identity of the aircraft or information about the flight. Due to the increase in flights, frequency spectrum management is an area that has received great attention since the voice communication band is becoming increasingly inefficient. This trend triggered the synergy of several organizations and regulatory authorities such as Eurocontrol, Eurocae,

and the European Union Aviation Safety Agency (EASA) to develop modern and safe communication infrastructures. The Single European Sky ATM Research (SESAR) initiative [1] works towards this direction, where one of its goals is to develop the L-Band Digital Aeronautical Communication System (LDACS) [2], [3], [4]. It is a cellular-like digital signaling system for flight guidance, A/A and A/G communications operating within the L-band frequency spectrum, namely 960–1164 MHz.

It has been highlighted that existing aviation communications should be replaced by modern technologies that address important drawbacks related to cyber-security and resilience [4]. The Controller-Pilot Data Link Communications (CPDL), which is a communication system that allows pilots and air traffic controllers to exchange text-based messages instead of relying solely on voice radio, is a major candidate to employ LDACS.

A. Contributions and structure

The goal of this paper is to perform an in-depth security evaluation of the LDACS communication system towards achieving high post-quantum cryptographic security. The key contributions of this work are summarized as follows:

- We analyze quantum, side-channel, and Machine Learning (ML) attacks that can reveal crucial secret information stored at the aircraft and ground control station radio units, such as PUF challenge and response attributes.
- We illustrate that large size of a PUF response is more difficult to be revealed, compared to small size responses.
- We propose a Public Key Infrastructure (PKI) relying on post-quantum cryptography as an authentication countermeasure, utilizing FALCON and Dilithium, as an efficient set up in respect to latency and power consumption.
- We suggest a CRYSTALS-Kyber hardware implementation that offers resistance against side-channel attacks and fast routine executions.

The rest of the paper is structured as follows. Section II presents a brief but comprehensive background of the ingredients of LDACS and also the concepts used to evaluate its security. Section III analyzes the methodology followed to evaluate the authentication and secrecy modules of LDACS and includes the strategic plan to benchmark the performance of the proposed PQC schemes. A discussion on the paper's

findings is presented in Section IV, whereas conclusions and future research directions are provided in Section V.

II. PRELIMINARIES

This section provides the necessary background to comprehend the physical unclonable functions and post-quantum cryptography algorithms that constitute the key security modules of LDACS. In addition, side-channel and information set decoding attacks are briefly analyzed, along with the fundamental process of registering a new radio unit to the LDACS infrastructure.

A. Quantum computing

Quantum computers [5] are the modern holy grail of computing. Although production is still in its early stages, large companies, such as IBM and Google, are developing powerful systems. Despite the computational power that quantum computers provide, quantum-based attacks that exploit this power rely on the quantum speedups achieved by few well-known algorithms. In 1994, Peter Shor developed a quantum algorithm that makes the prime integer factorization feasible [6]. The powerful and widely used Rivest–Shamir–Adleman (RSA) cryptographic scheme is based on this problem and is considered obsolete for the post-quantum era [7].

The security of RSA is not only the aspect threatened by quantum computers. Lov Grover introduced a quantum algorithm in 1996 that provides a quadratic speedup for search problems. Thus, while a brute force attempt requires $O(N)$ operations to find an element in an unsorted list of length N , Grover's algorithm needs only $O(\sqrt{N})$ operations [8]. Considering this, any attack on retrieving a cipher's key still needs exponential time. However, quadratic speedup may be introduced as part of sophisticated attacks, as shown below.

B. Post-quantum signature schemes

FALCON [9] and Dilithium [10] are some of NIST's PQC digital signature standards, based on lattices and depending on the hardness of finding short vectors in a lattice. Both are highly secure and designed to be small and fast. Performance and applicability benchmarks in [11], [12] indicate that Dilithium offers advantages in low power scenarios consuming less energy, while FALCON is superior in signature verification speed.

C. LDACS building blocks

Considering the computing power of quantum computers, the National Institute of Standards and Technology (NIST) launched a process to standardize quantum resistant Key Encapsulation Mechanisms (KEM) and Digital Signatures [13]. During the final round of the evaluation process, the Classic-McEliece and Bit Flipping Key Encapsulation (BIKE) algorithms were proposed as robust candidates for standardization. Although CRYSTALS-Kyber is the selected KEM scheme, the aforementioned two schemes are remarkable proposals, and the Classic-McEliece is also under standardization [14].

A Physical Unclonable Function (PUF) can be any physical object where, for a given input and condition, called *challenge*,

provides a physically deterministic output, called *response*; this deterministic attribute is utilized as *digital fingerprint* [15]. The delay that semiconductors remark, constitutes the reason that PUF behave as unique identifier for a hardware because every time someone uses the same challenge, at the same hardware, it always acts with the same delay and thus produces the same response.

D. LDACS registration of radio units

Many studies have been devoted on LDACS security [16], [17], [18], [19], focusing on different aspects. However, a recent work [20] aims to develop a lightweight system that reduces latency until a session key is established. According to this work, the *Registration* process must take place before a radio unit is serviceable and ready to be installed at an Aircraft Station (AS). During this process, both the AS and the Ground Station Controller (GSC) radio units exchange messages via a private and secure channel. This process causes the AS LDACS radio unit to create a PUF Challenge-Response-Pair (CRP) and send it to GSC's radio unit. Finally, the latter saves the CRP (C_{AS}, R_{AS}) at its non volatile memory, along with other information like the public key P_{AS} of KEM, the International Civil Aviation Organization Address ICAO_{AS} and the pseudo-address τ_{AS} , which is computed as

$$\tau_{AS} = h(\text{ICAO}_{AS} || R_{AS}) \quad (1)$$

where h is a secure hash function.

E. LDACS security threats

The basic logic behind the class of Information Set Decoding (ISD) attacks relies on algorithms that solve the decoding problem for linear codes. In particular, ISD is applied to analyze the security of code-based cryptographic schemes [21]. These algorithms exploit the structural properties of error-correcting codes and reveal the original message received from the erroneous codeword. There are several attack methods against code-based ciphers, which leverage and considerably extend Prange's algorithm [22] to yield highly efficient ISD algorithms.

Side-channel attacks (SCA) exploit physical phenomena that occur as a result of computations performed by modern microelectronic devices. For instance, tasks executed by microprocessors require time and power, emitting electromagnetic signals and occasionally sound. These observable characteristics are the source of unintended information leakage. Adversaries can convert these emissions into tools and reveal critical information about cryptographic operations and stored secret quantities (e.g. cryptographic keys) [23].

III. EVALUATION OF LDACS

Systems like CPDL can operate before take-off or even before the engine starts, sending automated messages to the GSC. By that, the control tower becomes aware of the flight readiness and provides the clearance to begin the preflight procedures. The LDACS Mutual Authentication and Key Establishment (MAKE) protocol starts communication by exchanging the messages M_1 , M_2 , M_3 and M_4 , as illustrated in

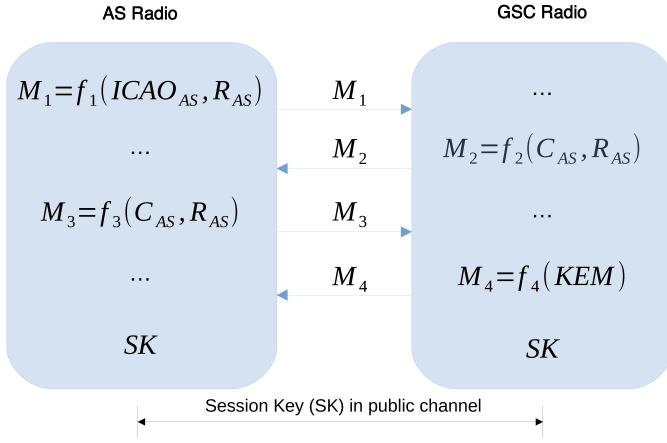


Fig. 1. LDACS lightweight MAKE protocol

Fig. 1 [20]. All messages contain information generated from several processes that are represented in compact form as f_1 , f_2 , f_3 and f_4 , respectively. So, an adversary may attempt to intercept the communication before an AS takes off. With the appropriate hardware, as mentioned in [24], it is feasible to tune the appropriate L-band frequency channel and receive these signals, giving the ability to disclose τ_{AS} from M_1 .

A. Authentication vulnerabilities

Several methodologies to disclose τ_{AS} , R_{AS} , and C_{AS} of the LDACS system are described below.

1) *Disclosure of τ_{AS}* : The AS acts as the client initiating the mutual authentication procedure by sending the message $M_1 = N_1 || \tau_{AS} || a$, where $a = h(N_1 || \tau_{AS})$. In [20] we can see that M_1 consists of a sequence of 48 bits for the header, 64 bits for the nonce N_1 , 24 bits for τ_{AS} , and 128 bits for a . By default, M_1 is not ciphered and an eavesdropper is able to distinguish the structure of this message, isolating τ_{AS} . Thus, in a small airport, where departures are infrequent, a threat actor is capable of recording τ_{AS} , matching it with the corresponding $ICA0_{AS}$.

2) *Disclosure of R_{AS}* : A recently proposed technology for LDACS considers a Static Random Access Memory (SRAM) PUF architecture [16], with a response length of 168 and 128 bits respectively [25], [26]. However, [27] introduces a new preimage quantum algorithm which extends Grover's algorithm and further reduces the cost from $O(2^{n/2})$ to $O(2^{n/3})$ operations. Under these circumstances and considering that $ICA0_{AS}$ is a public static attribute and R_{AS} a private static attribute during AS's lifetime, it is feasible to attack the aforementioned hash function by exploiting near-future quantum computing capabilities¹. Moreover, recent technologies such as the new *Willow*² quantum chip promise more computation capabilities, better error correction, and improved quantum performance. IBM foresees that the quantum computer *Blue*

TABLE I
TIME TO REVEAL R_{AS}

n	g	G	T (yrs)	g	G	T (yrs)
128	$5 \cdot 10^4$	10^{10}	1.1	$7 \cdot 10^5$	10^{12}	0.15
168	10^2	10^{10}	22	10^3	10^{12}	2.2
256	10^2	10^{10}	infeasible	10^2	10^{12}	58
512	10^2	10^{10}	infeasible	10^2	10^{12}	infeasible

*Jay*³ will have a billion gates, capable of executing an equivalent number of operations per second. Despite that Moore's law is not clearly applicable in quantum computing [28], we may take into account that there is a lot of effort to improve their capabilities. For example, according to IBM's provisions, quantum computers in 2025, 2027, 2029 and 2033 will have 5K, 10K, 100M and 1B gates respectively, which reflects an exponential-like growth rate. Based on the above, a quantum computer with one trillion gates by the end of the next decade seems to be feasible, in which case the quantum preimage algorithm will be able to extract R_{AS} from (1).

Let us denote by G the number of gates that are utilized per second, g the number of gates required per Grover step. Then, the upper bound of time T , according to [27], is calculated as

$$T = \frac{g}{G} 2^{n/3} \quad (2)$$

where the fraction g/G expresses the rate of execution time in seconds. Table I presents the execution time T needed for revealing the PUF response R_{AS} according to (2) for values of g adopted from [29], along with more optimistic ones.

Moreover, in [30] the authors state that the probability that a preimage attack succeeds is 0.63 using 2^l guesses, without prior knowledge, where l is the length of hash output. Considering that (1) has a 24-bit length, a classical computer would need around 16 million guesses to achieve this probability, which is rather feasible. The same study calculates a lower bound on the depth of quantum gates and the width of qubits required to apply the preimage attack on several hash functions. Indicatively, the authors focus on computing the cost of MD5 preimage for various search targets which are shown to be computationally affordable. For MD5, the cost is less than 30 million gates, enabling 801 qubits, which will be efficiently managed by Blue Jay.

In summary, R_{AS} is stored in both AS and GSC generated by the AS radio characteristics acting as its digital identity, while $ICA0_{AS}$ and R_{AS} remain unchanged. Hence, an adversary could attempt a partial attack against (1) trying to reveal AS's PUF secret values. Consequently, the disclosure of R_{AS} promises a high potential to further guess the challenge C_{AS} .

3) *Guessing of C_{AS}* : In connection with the definition of PUF, we note that PUF acts as an one-to-one mapping because any CRP is unique [15]. Since an one-to-one mapping is invertible, an adversary may attempt to find the challenge of

¹<https://spectrum.ieee.org/ibm-quantum-computer-2668978269>

²<https://blog.google/technology/research/google-willow-quantum-chip/>

³<https://www.ibm.com/roadmaps/quantum/2029/>

a PUF given the response. Although there is no mathematical expression available to quantify C_{AS} , R_{AS} , we can determine

$$C_{AS} \equiv \text{PUF}^{-1}(R_{AS})$$

by using ML models. The Covariance Matrix Adaptation Evolution Strategy (CMA-ES) is a powerful stochastic numerical optimizer to solve black box problems [31]; it was used to attack an Arbiter PUF (APUF) with remarkable success [32], where the precision in predicting the APUF response, given a challenge, reached 97%. Therefore, if an adversary attempts to guess C_{AS} , given R_{AS} , he must acquire the real SRAM-PUFs coming from the same manufacturing batch as they may have a correlation and the Hamming distance between responses can be lower than two completely unrelated chips. Alternately, if an attacker can precisely replicate a chip using advanced fabrication techniques, two SRAM-PUFs could be nearly identical. Even though two SRAM-PUFs will not be identical, a well-trained ML model might still predict another SRAM-PUF's responses with a certain probability.

In any case, due to the fact that an ML model can learn common response biases or structural correlations, a realistic CRP data set can be built if responses from multiple SRAM-PUFs are gathered. For the sake of the experiment, aligned with the logic of [32], a theoretical data set was constructed, with random 32-bit challenges, affected by micro-noise⁴. Then, the CMA-ES algorithm would need a dataset of 100 CRPs to guess the challenge in 6 seconds with precision 98%.

4) *Disclosure of C_{AS} , R_{AS} from GSC*: An outstanding type of SCA attack was given in [35] targeting typical workstations. Regardless of whether a computer is networked, stand-alone, or air-gaped the proposed attack can use the Serial Advanced Technology Attachment (SATA) cable as a wireless antenna to transfer data as radio signals. SATA is a bus interface widely used in modern computers and connects the host bus to mass storage devices such as hard-disk and solid-state drives. The results demonstrate that SATA bus electromagnetic signals can lead to significant information leakage.

While not many details are revealed regarding the LDACS ground-based hardware architecture, there is clear assumption that a permanent drive should store all the data related to each AS assigned to this service. So, when the GSC computer receives M_1 , it immediately locates the value τ_{AS} and loads the corresponding CRP from a database. The logic of this attack is to match M_1 timestamps, thus τ_{AS} (or even better ICAO_{AS}) to pairs (C_{AS}, R_{AS}) obtained from the GSC computer's drive.

5) *PUF aging*: Aging refers to the gradual degradation of semiconductor device performance over time due to various physical mechanisms and can cause changes in the transistors' behavior, leading to variations in circuit performance [33]. Two primary aging effects are:

- **Bias Temperature Instability (BTI)**: This phenomenon affects PUF transistors when subjected to prolonged voltage stress at elevated temperatures.

⁴Avionics systems operate in almost ideal environment. Countermeasures are taken for the interference of temperature and electromagnetic emissions

- **Hot Carrier Injection (HCI)**: HCI occurs when high-energy carriers gain sufficient energy to overcome potential barriers at PUF transistors.

In the context of LDACS, as PUF responses may become less stable, there is an increased risk for the authentication mechanism to fail. The number of bits flipping in SRAM-PUFs was shown to increase by 19.3% over two years of usage [34].

B. Secrecy vulnerabilities

A brief analysis of KEMs proposed in the literature about the LDACS system is next described.

1) *McEliece algorithm*: The use of the Classic-McEliece as a KEM in LDACS system has been recently proposed [17]. The McEliece cryptosystem achieves security under chosen-plaintext attacks relying on the hardness of decoding random linear codes [36]. It uses binary Goppa codes that are hard to decode without the secret key but still have efficient decoding algorithms for the legitimate users. Although numerous attack strategies have been proposed, the McEliece cryptosystem has demonstrated exceptional resilience and stability.

Attacks: A state-of-the-art ISD attack against McEliece has been thoroughly analyzed in [37]. This class of attacks are performed with classical computers. However, that kind of attacks may become more efficient when quantum computers contribute to resolve search sub-problems. A notable paradigm is the quantum information-set decoding, proposed in [38]. Moreover, another approach that combines SCA, ISD, and ML is able to reveal the secret key, at all security levels [39].

2) *CRYSTALS-Kyber algorithm*: The 512/768 versions of the algorithm have been proposed as a KEM for LDACS [18]. The CRYSTALS-Kyber is a lattice-based cryptographic scheme designed to provide security against attacks where adversaries have access to plaintext and ciphertext pairs [40]. Additionally, the cipher is computationally efficient and has small key sizes compared to other post-quantum schemes. It is well-suited for environments such as embedded systems and IoT devices.

Attacks: Advanced SCA poses challenges for cryptographic systems, even those with highly robust security. Several studies highlight persistent vulnerabilities, even for the CRYSTALS-Kyber standard by NIST [41].

3) *BIKE algorithm*: The BIKE cipher has been proposed for the establishment of the session key in LDACS [19], [20]. Designed to provide secure key encapsulation, BIKE is also relying on the NP-hardness of decoding problems using quasi-cyclic codes. It is one of the code-based cryptographic schemes that made it to the final stage of the NIST PQC standardization process. Although not selected for standardization, BIKE is a notable alternative for quantum-resistant cryptography.

Attacks: The software implementations of BIKE can expose sensitive information. This KEM is vulnerable by combining SCA (power analysis) and ISD attacks, allowing the complete private key to be extracted even from one power consumption trace, during the decapsulation phase [42], [43].

TABLE II
QUANTUM RESISTANT KEMS

NIST	BIKE	CRYSTALS-Kyber	Classic-McEliece
Level 3	BIKE-L3	CRYSTALS-Kyber768	460896
Level 5	BIKE-L5	CRYSTALS-Kyber1024	6960119

C. PQC performance evaluation

Benchmarks on the time required to execute encapsulation, decapsulation, and symmetric cipher operations for the above KEMs are next provided.

Classic-McEliece and BIKE are more vulnerable combining SCA and ISD attacks, rather than kyber that suffers solely by SCA. To extend comparisons between the KEMs of Table II, performance benchmarks are also included. The results in Fig. 2 indicate that CRYSTALS-Kyber is the fastest cipher with an execution time below 150 *ms*. The AES-256-CBC was used as session cipher during our experiments using the key of the encapsulation and decapsulation mechanism. Fig. 3 illustrates that ciphers' workload depends on the KEM's generated session key; more precisely, the keys generated by CRYSTALS-Kyber spend less time during the symmetric encryption and decryption process. Since entropy is the information-theoretic measure that quantifies the uncertainty of ciphertext, higher values indicate that less patterns are included in the ciphertext increasing uncertainty and attacks' complexity. Fig. 4 shows entropy comparisons between ciphertexts generated by AES-256-GCM with the symmetric key generated from the encapsulation mechanism. It is clear that symmetric keys generated from BIKE-L3/5 have the worst entropy performance, whereas the other KEMs include a higher and more stable uncertainty in their ciphertext.

IV. DISCUSSION

During the evaluation of LDACS in Section III, it becomes evident that the proposed PUF-based authentication protocol may leak crucial secret information. Instead, and with regard to the proposal for A/A communications using LDACS [44], digital certificates and certificate authorities could be implemented for LDACS [18]. PKI in aviation has already been studied in [45], mentioned for LDACS in [46] and thus could be developed in this domain, while this aviation communication protocol has not yet been standardized, in terms of information security that is expected to serve the aviation industry for the next decades.

As seen in Section III-B, McEliece, CRYSTALS-Kyber, and BIKE are fast and solid quantum-resistant KEM candidates for LDACS, but the architecture of the MAKE protocol to keep the public key secret, as highlighted in Section II, diverts the effort from the PKI mindset, which is an established architecture in the vast majority of security applications and protocols. However, keeping the AS's public key secret degenerates the fundamental principle of public key cryptography. Nonetheless, to ensure long-term security and resilience against quantum threats, LDACS should adopt a PQC-based

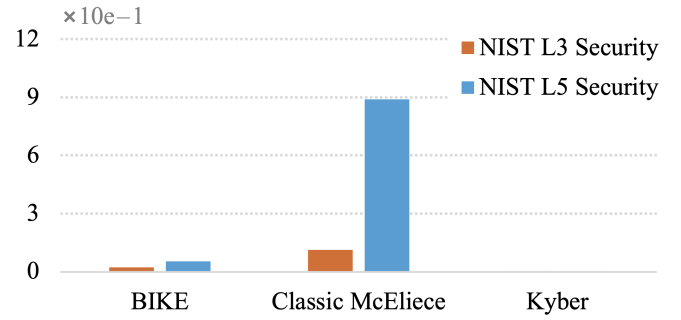


Fig. 2. KEM Benchmark: Encapsulation and decapsulation time in seconds

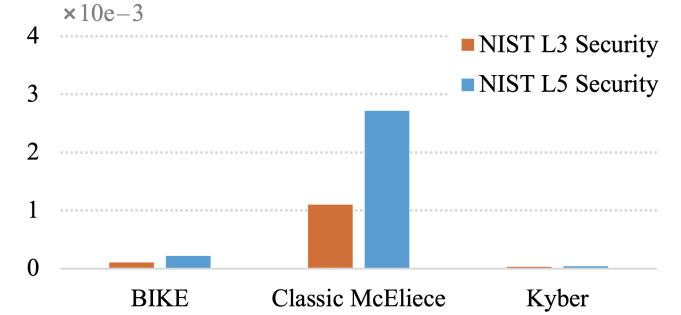


Fig. 3. AES-256-CBC Encryption+Decryption Time, as $f(KEM)$

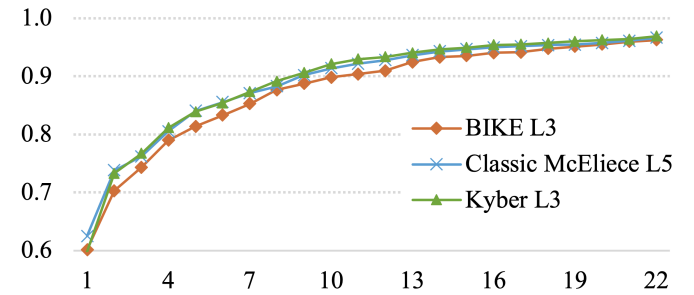


Fig. 4. AES-256-GCM ciphertext entropy after encryption, as $f(KEM)$

PKI since the very beginning, integrating FALCON-1024 and Dilithium for digital signatures, and CRYSTALS-Kyber for key exchange, as illustrated in Fig. 5. This proactive approach not only mitigates the risk of future quantum-based attacks but also ensures compatibility with evolving aviation security standards. Leveraging on FALCON's compact and efficient signatures for authentication, Dilithium's low-energy performance, which is crucial in constrained environments like the sensitive sector of aviation and CRYSTALS-Kyber's robust key encapsulation for secure communication, LDACS can establish a cryptographic foundation that remains secure throughout its operational lifetime. Consequently, it would be more convenient for a radio unit to start its service utilizing only PQC, avoiding the need for additional reconfiguration.

Concerning the latency and security of KEM, it is true that the more messages exchanged to establish a session key, the slower the system becomes, which is considered a serious

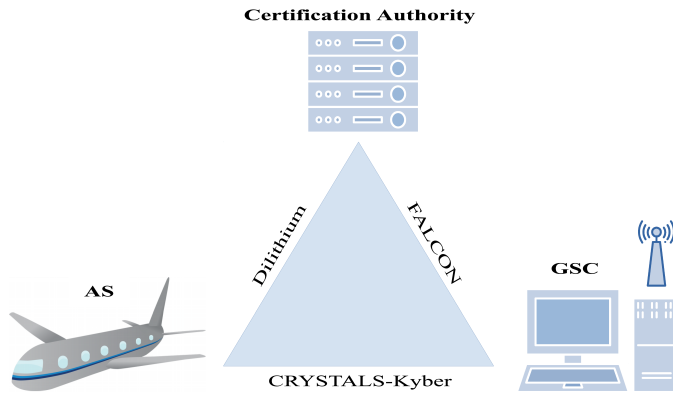


Fig. 5. The proposed PQC-based PKI approach

drawback in aviation; thus, researchers struggle to develop a highly lightweight MAKE protocol. In any case, regardless the adoption of PKI in LDACS architecture, a fast enough KEM scheme would reduce latency.

Since aviation is a critical sector, possible failures during authentication should be handled by redundant equipment, the integration of which increases the infrastructure's complexity. Such redundancy can be achieved by either a separate PUF circuit or a separate radio unit. The former approach requires specific handling synchronization procedures, due to the different CRP, between the AS and GSC. On the other hand, the latter approach demands more maintenance such as removing the radio unit from the aircraft and re-registering at a safe environment.

In contrast, a PKI-based architecture would increase flexibility, allowing a redundant software implementation to perform the cryptographic executions using the key pair generated by the hardware implementation of the KEM. Moreover, if asymmetric radio keys are dynamically renewed, this could be done as ground maintenance procedures, without removing the radio. Furthermore, if the development still consists of PUF, then the possibility of increased latency may occur when larger response bit sequence is introduced to avoid III-A2 quantum attacks.

Regarding CRYSTALS-Kyber, our experiments showed that it is the fastest and most secure KEM for LDACS, making it a strong candidate for future deployments. For example, a software implementation of CRYSTALS-Kyber runs in almost 130 ms at security level-5 as shown in Fig. 2. Furthermore, there are hardware implementations that improve its protection against SCA [47], [48], and additionally reduce the run time in almost 580 μ s. This implementation improves the processing speed by 99% while simultaneously mitigating electromagnetic leakage. In this context, the focus should be on reducing the emissions produced during functions like decryption, while strengthening its resistance against SCA to create a more resilient, fast and perpetuating KEM customized for crucial aeronautical communications.

As aviation relies on well-established standards and practices, where a common terminology is a key factor in com-

munications, this may impact the session key's security. The messages being sent through the CPDL are standardized to be as short and meaningful as possible [49]. Due to this approach, these messages have reserved words and phrases like

“REQUEST CLIMB FL350” and
“ACCEPT, REPORT ON FL350”.

When repetitively ciphered by the same session key, the SCA can reveal part of this key, which is known as *known-plaintext attacks*, a concept that must be highly considered. In relation to the session key, should this be renewed often to shield it against relevant attacks, CRYSTALS-Kyber, as the faster KEM, would be once more a prominent choice.

V. CONCLUSIONS AND FUTURE WORK

The paper presented an evaluation of LDACS post-quantum security focusing on the authentication mechanisms. Although PUFs are a state-of-the-art technology for authentication, our analysis demonstrates that an adversary could extract secret attributes, enabling impersonation attacks in future quantum-capable threat models. Moreover, PUF aging could introduce reliability issues, potentially impacting the establishment of authentication procedures. To address this vulnerability, a PQC-based PKI for LDACS authentication was proposed with CRYSTALS-Kyber to thrive in performance and security.

To allow reproducibility and encourage further research, the implementations of the evaluations reported in the paper are available on GitHub⁵. The code developed to generate data for Figs. 2, 3, 4 utilizes the Python wrapper of the Open Quantum Safe library. The CMA-ES ML algorithm used to predict the PUF challenge is generated by a Python script, based on the logic of the related reference literature. All these scripts have been executed on an M2 processor with 16GB RAM.

Future work will focus on evaluating the proposed implementations in real-world conditions, including practical deployment scenarios and adversarial testing. Moreover, more research is needed to optimize power efficiency and explore the integration of the PQC algorithms tailored for LDACS. Investigating countermeasures against advanced SCA, such as threat analysis based on deep learning, remains an open research direction.

REFERENCES

- [1] R. Koelle and M. Hawley, “Sesar security 2020: How to embed and assure security in system-of-systems engineering?”, In *proc. 2012 Integrated Communications, Navigation and Surveillance Conference*, pp. E8-1-E8-11, 2012.
- [2] C. Rihacek, B. Haindl, P. Fantappie, S. Pierattelli, T. Gräupl, M. Schnell, and N. Fistas, “L-band digital aeronautical communications system (LDACS) activities in SESAR2020”, In *proc. 2018 Integrated Communications, Navigation, Surveillance Conference*, pp. 4A1-1-4A1-8, 2018.
- [3] N. Mäurer, T. Gräupl, C. Gentsch, T. Guggemos, M. Tiepelt, C. Schmitt, and G.D. Rodosek, “A secure cell-attachment procedure of LDACS”, In *proc. 2021 IEEE European Symposium on Security and Privacy Workshops*, pp. 113–122, 2021.
- [4] N. Mäurer, T. Gräupl, and C. Schmitt, “L-band digital aeronautical communications system (LDACS)”, RFC 9372, 2023.

⁵<https://github.com/konnnGit/LDACS-Security-Evaluation>

- [5] B. Zohuri and F. Mossavar-Rahmani, "What is quantum computing and how it works", *Journal of Material Sciences & Manufacturing Research*, pp. 1–5, 2020.
- [6] H. Wong, "Shor's algorithm", In *Introduction to Quantum Computing*, pp. 279–288, Springer, 2022.
- [7] V. Bhatia and K.R. Ramkumar, "An efficient quantum computing technique for cracking RSA using Shor's algorithm", In *proc. 2020 IEEE 5th International Conference on Computing Communication and Automation*, pp. 89–94, 2020.
- [8] A. Mandviwalla, K. Ohshiro, and B. Ji, "Implementing Grover's algorithm on the IBM quantum computers", In *proc. 2018 IEEE International Conference on Big Data*, pp. 2531–2537, 2018.
- [9] D. Soni, K. Basu, M. Nabeel, N. Aaraj, M. Manzano, and R. Karri, "FALCON", in *Hardware Architectures for Post-Quantum Digital Signature Schemes*, pp. 31–41, 2021.
- [10] Ducas, Léo, et al. "CRYSTALS-Dilithium: A Lattice-Based Digital Signature Scheme.", *IACR Trans. Cryptogr. Hardw. Embed. Syst.* 2018.1 (2018): 238–268.
- [11] Vidaković, Marin & Kruno, Milicevic. (2023). "Performance and Applicability of Post-Quantum Digital Signature Algorithms in Resource-Constrained Environments", *Algorithms*. 16. 518.
- [12] Maximilian Pursche and Nikolai Puch and Sebastian N. Peters and Michael P. Heintl, "SoK: The Engineer's Guide to Post-Quantum Cryptography for Embedded Devices", published in *Cryptology ePrint Archive*, Paper 2024/1345, 2024
- [13] L. Chen, L. Chen, S. Jordan, Y.-K. Liu, D. Moody, R. Peralta, R. Perlner, and D. Smith-Tone, "Report on post-quantum cryptography", National Institute of Standards, Technology, 2016.
- [14] R. Bavdekar, E. Jayant Chopde, A. Agrawal, A. Bhatia, and K. Tiwari, "Post quantum cryptography: A review of techniques, challenges and standardizations", In *proc. 2023 International Conference on Information Networking*, pp. 146–151, 2023.
- [15] C. Herder, M.-D. Yu, F. Koushanfar, and S. Devadas, "Physical unclonable functions and applications: A tutorial", *Proceedings of the IEEE*, vol. 102, no. 8, pp. 1126–1141, 2014.
- [16] N. Mürer, T. Gräupl, C. Schmitt, and G.D. Rodosek, "PMAKE: Physical unclonable function-based mutual authentication key exchange scheme for digital aeronautical communications", In *proc. 2021 IFIP/IEEE International Symposium on Integrated Network Management*, pp. 206–214, 2021.
- [17] D.M. Mielke, N. Mürer, T. Gräupl, and M.A. Bellido-Manganell, "Quantum Applications - Fachbeitrag: Getting Civil Aviation Ready for the Post Quantum Age with LDACS," *Digitale Welt*, vol. 5, pp. 28–33, 2021.
- [18] N. Mürer, T. Gräupl, and C. Schmitt, "Efficient control-channel security for the aeronautical communications system LDACS", In *proc. 2023 IEEE 24th International Symposium on a World of Wireless, Mobile and Multimedia Networks*, pp. 407–412, 2023.
- [19] M.R. Nosouhi, S.W.A. Shah, L. Pan, and R. Doss, "Bit flipping key encapsulation for the post-quantum era", *IEEE Access*, vol. 11, pp. 56181–56195, 2023.
- [20] S. Khan, G.S. Gaba, A. Gurtov, N. Mürer, T. Gräupl, and C. Schmitt, "Enhancing cybersecurity for LDACS: A secure and lightweight mutual authentication and key agreement protocol", In *proc. 2023 IEEE/AIAA 42nd Digital Avionics Systems Conference*, pp. 1–10, 2023.
- [21] J.T. Coffey and R.M. Goodman, "The complexity of information set decoding", *IEEE Transactions on Information Theory*, vol. 36, no. 5, pp. 1031–1037, 1990.
- [22] E. Prange, "The use of information sets in decoding cyclic codes", *IEEE Transactions on Information Theory*, vol. 8, no. 5, pp. 5–9, 1962.
- [23] F.-X. Standaert, "Introduction to side-channel attacks", In *proc. Secure Integrated Circuits and Systems*. Springer, pp. 27–42, 2010.
- [24] E. Orye, G. Visky, and O. Maennel, "Analysing the actual use of controller-pilot data link communications", *Engineering Proceedings*, vol. 28, no. 1, pp. 18, 2022.
- [25] S. Eiroa, J. Castro, M.C. Martínez-Rodríguez, E. Tena, P. Brox and I. Baturone, "Reducing bit flipping problems in SRAM physical unclonable functions for chip identification", in *proc. 2012 19th IEEE International Conference on Electronics, Circuits, and Systems*, pp. 392–395, 2012.
- [26] A. Miller, Y. Shifman, Y. Weizman, O. Keren and J. Shor, "A highly reliable SRAM PUF with a capacitive preselection mechanism and pre-ECC BER of 7.4×10^{-10} ", in *proc. 2019 IEEE Custom Integrated Circuits Conference*, pp. 1–4, 2019.
- [27] P. Wang, S. Tian, Z. Sun, and N. Xie, "Quantum algorithms for hash preimage attacks", *Quantum Engineering*, vol. 2, no. 2, pp. e36, 2020.
- [28] Olivier Ezratty, "Is there a Moore's law for quantum computing?", *arxiv.org* 2023.
- [29] Ramos-Calderer, S., Bellini, E., Latorre, J.I. et al. "Quantum search for scaled hash function preimages". *Quantum Inf Process* 20, 180 (2021).
- [30] R. H. Preston, "Applying Grover's Algorithm to Hash Functions: A Software Perspective", in *IEEE Transactions on Quantum Engineering*, vol. 3, pp. 1–10, 2022, Art no. 2500710, doi: 10.1109/TQE.2022.3233526.
- [31] K. Varelas, A. Auger, D. Brockhoff, N. Hansen, O.A. ElHara, Y. Semet, R. Kassab, and F. Barbaresco, "A comparative study of large-scale variants of CMA-ES", In *proc. Parallel Problem Solving from Nature*. Springer LNCS, vol. 11101, pp. 3–15, 2018.
- [32] C. Xu, L. Zhang, M.-K. Law, X. Zhao, P.-I. Mak and R.P. Martins, "Modeling-attack-resistant strong PUF exploiting stagewise obfuscated interconnections with improved reliability", *IEEE Internet of Things Journal*, vol. 10, no. 18, pp. 16300–16315, 2023.
- [33] Bhatta, N.P., Singh, H., Ghimire, A. et al. "Analyzing Aging Effects on SRAM PUFs: Implications for Security and Reliability". *J Hardw Syst Secur* 8, 174–186 (2024).
- [34] Wang, Rui, et al. "Long-Term Continuous Assessment of SRAM PUF and Source of Random Numbers.", *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, IEEE, 2020.
- [35] M. Guri, "SATAn: Air-gap exfiltration attack via radio signals from SATA cables", In *proc. 2022 19th Annual International Conference on Privacy, Security & Trust*, pp. 1–10, 2022.
- [36] D.J. Bernstein, T. Chou, T. Lange, I. von Maurich, R. Misoczki, R. Niederhagen, E. Persichetti, C. Peters, P. Schwabe, N. Sendrier, J. Szefer, and W. Wang, "Classic McEliece: Conservative code-based cryptography", Technical Report, Nov. 2017.
- [37] D.J. Bernstein, T. Lange, and C. Peters, "Smaller decoding exponents: Ball-collision decoding", In *proc. Advances in Cryptology — CRYPTO 2011*. Springer LNCS, vol. 6841, pp. 743–760, 2011.
- [38] D.J. Bernstein, "Grover vs. McEliece", in *proc. Post-Quantum Cryptography — PQCrypto 2010*. Springer LNCS, vol. 6061, pp. 73–80, 2010.
- [39] Q. Guo, A. Johansson, and T. Johansson, "A key-recovery side-channel attack on classic McEliece implementations", *IACR Transactions on Cryptographic Hardware and Embedded Systems*, vol. 2022, no. 4, pp. 800–827, 2022.
- [40] J. Bos, L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, J. M. Schanck, P. Schwabe, G. Seiler, and D. Stehlé, "CRYSTALS - Kyber: A CCA-secure module-lattice-based KEM", in *proc. 2018 IEEE European Symposium on Security and Privacy*, pp. 353–367, 2018.
- [41] M. Iavich and T. Kuchukhidze, "Investigating CRYSTALS-Kyber vulnerabilities: Attack analysis and mitigation", *Cryptography*, vol. 8, no. 2, 2024.
- [42] A. Cherie, N. Aragon, T. Richmond, and B. Gérard, "BIKE key-recovery: Combining power consumption analysis and information-set decoding", In *proc. Applied Cryptography and Network Security — ACNS 2023*. Springer LNCS, vol. 13905, pp. 725–748, 2023.
- [43] A.-L. Horlemann, S. Puchinger, J. Renner, T. Schamberger, and A. Wachter-Zeh, "Information-set decoding with hints", *Code-Based Cryptography — CBCrypto 2021*. Springer LNCS, vol. 13150, pp. 60–83, 2022.
- [44] M.A. Bellido-Manganell and M. Schnell, "Towards modern air-to-air communications: The LDACS A2A mode", in *proc. 2019 IEEE/AIAA 38th Digital Avionics Systems Conference*, pp. 1–10, 2019.
- [45] B. Crowe, "Proposed AeroMACS PKI specification is a model for global and national aeronautical PKI deployments", in *proc. 2016 Integrated Communications Navigation and Surveillance*, pp. 1–19, 2016.
- [46] L.J.A. Jansen, N. Mürer, and T. Gräupl, "Secure point-to-point long-distance multi-hop connections in a dense airplane mesh-network using LDACS", in *proc. 2022 IEEE/AIAA 41st Digital Avionics Systems Conference*, pp. 1–10, 2022.
- [47] Y. Ji, R. Wang, K. Ngo, E. Dubrova and L. Backlund, "A side-channel attack on a hardware implementation of CRYSTALS-Kyber", in *proc. 2023 IEEE European Test Symposium*, pp. 1–5, 2023.
- [48] A. Jati, N. Gupta, A. Chattopadhyay, and S.K. Sanadhya, "A configurable CRYSTALS-Kyber hardware implementation with side-channel protection", *ACM Transactions on Embedded Computing Systems*, vol. 23, no. 2, article 33, pp. 1–25, 2024.
- [49] International Civil Aviation Organization, *Global Operational Data Link (GOLD) Manual*, Document 10037, 1st ed., 2017.